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CS82A - Data Science

Text Book Problem 4.4

Temperature Categories

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Writing a python script for each day's temperature requires a **for loop** and **if/elif/else clause.** When we limit the code to an if/elif/else clause we return results for only the last index of the array 30. Incorporating a for loop allows iterations across each value of the list, Temperatures. Since we are not sure if "every value labeled and boundaries behave as the stated cutoffs promise" we make a second list which holds a range of values from -1 to 50 called Temperatures_2. According to our textbook problem 4.4, categorize each temperature from the list Temperatures, as:

- "Cold" if Temperature is below 15 Celcius. "temp < 15"

- "Warm" if Temperature is between 15 and 25 Celcius inclusive. "temp <= 25 and temp >= 15"

- "Hot" if Temperature is above 25 Celcius. "temp > 25"

- Temperatures = [10, 16, 23, 28, 14, 22, 30]

Next we can place our code into a function called temperature_function() and run a range of values from -1 to 50 to make sure the boundaries are as stated.

The function lets us re-use our code on a new list. We placed our printing statements inside the function, because printing the values of a list from outside the function returns the first value of the list only. We called the function with different lists to observe results.

In [1]:

```python
Temperatures =[ 10,16,23,28,14,22,30] # Celcius temps.
Temperatures_2 =[-1,0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,
# 1-14 cold
# 15-25 warm
# 26- hot
def temperature_function(Temperatures):
    print("The Temperatures of the Week (Celcius): ")
    print("-"*40)
    for each_temperature in Temperatures:
        if each_temperature < 15:
            print(each_temperature,"Cold")
        elif each_temperature <= 25:
            print(each_temperature,"Warm")
        else:
            print(each_temperature,"Hot")
temperature_function(Temperatures)
print("\n")
temperature_function(Temperatures_2)
```

```
The Temperatures of the Week (Celcius):
----------------------------------------
10 Cold
16 Warm
23 Warm
28 Hot
14 Cold
22 Warm
30 Hot
```

```
The Temperatures of the Week (Celcius):
----------------------------------------
-1 Cold
0 Cold
1 Cold
2 Cold
3 Cold
4 Cold
5 Cold
6 Cold
7 Cold
8 Cold
9 Cold
10 Cold
11 Cold
12 Cold
13 Cold
14 Cold
15 Warm
16 Warm
17 Warm
18 Warm
19 Warm
20 Warm
21 Warm
22 Warm
23 Warm
24 Warm
25 Warm
26 Hot
27 Hot
28 Hot
29 Hot
30 Hot
31 Hot
32 Hot
```

```
33 Hot
34 Hot
35 Hot
36 Hot
37 Hot
38 Hot
39 Hot
40 Hot
41 Hot
42 Hot
43 Hot
44 Hot
45 Hot
46 Hot
47 Hot
48 Hot
49 Hot
50 Hot
```

The output displays the values from -1 to 14 labeled as Cold, values from 15 to 25 labeled as Warm, and 26 + labeled as Hot. The heading reads "The Temperatures of the Week (Celcius):"

End Of Lab - Behzad Ghabaei